

Solutions - MIE230 More Interesting Problems

Barsam Rahimi

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Problem 1 - Solution

The two determinants evaluate as:

$$\det \begin{pmatrix} x & 0 \\ 0 & -y^2 \end{pmatrix} = -xy^2, \quad \det \begin{pmatrix} e^{-x^2} & -1 \\ 1 & e^{-y^2} \end{pmatrix} = e^{-(x^2+y^2)} + 1$$

So the spinor field is:

$$\sigma = \left(-Axy^2, B \left(3 - i \left(e^{-(x^2+y^2)} + 1 \right) \right) \right)$$

with magnitude:

$$|\sigma| = \sqrt{A^2x^2y^4 + B^2 \left(9 + \left(e^{-(x^2+y^2)} + 1 \right)^2 \right)}$$

The normalization condition requires:

$$\iint_{\mathbb{R}^2} \left(A^2x^2y^4 + B^2 \left(9 + \left(e^{-(x^2+y^2)} + 1 \right)^2 \right) \right)^{1/4} dA = 1$$

For any $A \neq 0$, the contribution of the first term alone gives:

$$|A|^{1/2} \iint_{\mathbb{R}^2} |x|^{1/2} |y| dA = |A|^{1/2} \left(\int_{\mathbb{R}} |x|^{1/2} dx \right) \left(\int_{\mathbb{R}} |y| dy \right) = \infty$$

since neither $|x|^{1/2}$ nor $|y|$ is integrable over $\mathbb{R}$. This forces $A = 0$. With $A = 0$, converting to polar coordinates $r^2 = x^2 + y^2$ gives:

$$2\pi B^{1/2} \int_0^\infty \left(9 + \left(e^{-r^2} + 1 \right)^2 \right)^{1/4} r dr = 1$$

As $r \rightarrow \infty$, the integrand behaves like $10^{1/4} \cdot r$, so the integral diverges for any finite $B \neq 0$.

Therefore, there exist no finite values of A and B that normalize this distribution.

Physical remark: This mirrors the non-normalizability of plane wave solutions over unbounded domains. A physically realizable quantum state must decay sufficiently fast at infinity, a condition this spinor field does not satisfy.

Problem Insight

The trap in this problem is algebraic: students will attempt to solve for A and B assuming a finite normalization integral exists. The key realization is that the first spinor component produces a polynomial term $x^2 y^4$ under the integral over $\mathbb{R}^2$, which diverges regardless of how small A is made — as long as $A \neq 0$. The second component's integral also diverges independently. Therefore, no finite values of A and B can normalize this distribution.

Problem 2 - Solution

A natural parametrization of a helix on $M : z = x^2 + y^2$ takes $x = r \cos t$, $y = r \sin t$, giving $z = r^2$:

$$r_1(t) = [r \cos t, r \sin t, r^2]$$

Imposing unit speed:

$$r'_1(t) = [-r \sin t, r \cos t, 0] \implies |r'_1(t)| = r = 1$$

So the curve is:

$$r_1(t) = [\cos t, \sin t, 1]$$

After $t = 12$ s the particle follows a straight line, so the most natural basis for analyzing its trajectory is the Frenet-Serret frame inherited from the helix at $t = 12$. The unit tangent is:

$$\mathbf{T} = r'_1(t)|_{t=12} = [-\sin(12), \cos(12), 0]$$

The principal normal is:

$$\mathbf{N} = \frac{r_1''(t)}{|r_1''(t)|} \Big|_{t=12} = [-\cos(12), -\sin(12), 0]$$

And the binormal:

$$\mathbf{B} = \mathbf{T} \times \mathbf{N} = [0, 0, 1]$$

Since the particle travels in the direction of $\mathbf{T}$ along a straight line, $\{\mathbf{T}, \mathbf{N}, \mathbf{B}\}$ form the natural basis. Note that $\mathbf{N}$ and $\mathbf{B}$ are not uniquely determined by the straight-line trajectory alone (zero curvature), which is precisely why inheriting the TNB frame from the helix at $t = 12$ is the natural choice.

Since r_1 was parametrized with unit speed, the total arc length of the helical trajectory is:

$$L = \int_0^{12} |r_1'(t)| dt = \int_0^{12} 1 dt = 12 \text{ units}$$

Problem Insight

The unit speed condition immediately fixes $r = 1$, collapsing what looks like a family of helices to a unique curve. The subtlety of the problem is found in the Frenet frame after $t = 12$: a straight-line trajectory has zero curvature, so $\mathbf{N}$ and $\mathbf{B}$ are undefined from the new path alone. Students must recognize that the physically natural choice is to inherit the frame continuously from the helical phase. The arc length calculation is then trivial by construction.

Problem 3 - Solution

Let $I = \int_{-\infty}^{\infty} e^{-\alpha x^2} dx$. Squaring and combining into a double integral:

$$I^2 = \left(\int_{-\infty}^{\infty} e^{-\alpha x^2} dx \right) \left(\int_{-\infty}^{\infty} e^{-\alpha y^2} dy \right) = \iint_{\mathbb{R}^2} e^{-\alpha(x^2+y^2)} dA$$

Converting to polar coordinates:

$$I^2 = \int_0^{2\pi} \int_0^{\infty} e^{-\alpha r^2} r dr d\theta = 2\pi \int_0^{\infty} e^{-\alpha r^2} r dr$$

Substituting $u = \alpha r^2$, $du = 2\alpha r dr$:

$$I^2 = \frac{2\pi}{2\alpha} \int_0^\infty e^{-u} du = \frac{\pi}{\alpha}$$

Therefore:

$$I = \sqrt{\frac{\pi}{\alpha}}, \quad \alpha > 0$$

Problem Insight

The integral is not directly computable in one dimension. The trick is to square it and interpret the product as a double integral over $\mathbb{R}^2$, where the rotational symmetry of $e^{-\alpha(x^2+y^2)}$ makes polar coordinates natural. The $r dr$ factor then produces a simple exponential integral. Students who have not seen this before will find no obvious path forward — the insight is entirely in recognizing that a harder-looking 2D problem is actually easier.

Problem 4 - Solution

The volume of a swept solid is the product of the cross-sectional area and the arc length of the path, since the cross-section is always perpendicular to the curve:

$$V = A \cdot L$$

Computing the arc length of $r(t) = [2 \sin t, 2 \cos t, 3t]$ over $t \in [0, 4\pi]$:

$$r'(t) = [2 \cos t, -2 \sin t, 3] \implies |r'(t)| = \sqrt{4 \cos^2 t + 4 \sin^2 t + 9} = \sqrt{13}$$

$$L = \int_0^{4\pi} \sqrt{13} dt = 4\pi\sqrt{13}$$

The cross-sectional area for diameter $D = 3$, i.e. radius $\rho = 3/2$:

$$A = \pi\rho^2 = \frac{9\pi}{4}$$

Therefore the volume is:

$$V = \frac{9\pi}{4} \cdot 4\pi\sqrt{13} = 9\pi^2\sqrt{13}$$

Problem Insight

The key insight is recognizing that the swept volume formula $V = A \cdot L$ applies directly here, as CAD sweep operations maintain a perpendicular cross-section along the path by definition. Students who attempt to set up a full triple integral will find themselves in serious difficulty. The helix has a constant speed $\sqrt{13}$, which makes the arc length computation clean. The problem is fundamentally about recognizing the right geometric framework before doing any calculus.

Problem 5 - Solution

The acceleration of a fluid particle is the material derivative of the velocity field. For steady flow this reduces to:

$$\mathbf{a} = (\mathbf{v} \cdot \nabla)\mathbf{v}$$

whose components are $a_i = u\partial_x v_i + v\partial_y v_i + w\partial_z v_i$, evaluated at (x_0, y_0, z_0) . The partial derivatives of each velocity component are:

$$\partial_x u = 2x_0 y_0^3, \quad \partial_y u = 3x_0^2 y_0^2, \quad \partial_z u = 4z_0^3$$

$$\partial_x v = \cos x_0, \quad \partial_y v = \sin y_0 \cos z_0, \quad \partial_z v = \cos y_0 \sin z_0$$

$$\partial_x w = y_0 z_0 - e^{x_0+y_0+z_0}, \quad \partial_y w = x_0 z_0 - e^{x_0+y_0+z_0}, \quad \partial_z w = x_0 y_0 - e^{x_0+y_0+z_0}$$

giving acceleration components:

$$a_1 = u_0(2x_0 y_0^3) + v_0(3x_0^2 y_0^2) + w_0(4z_0^3)$$

$$a_2 = u_0 \cos x_0 + v_0 \sin y_0 \cos z_0 + w_0 \cos y_0 \sin z_0$$

$$a_3 = u_0(y_0 z_0 - e^{x_0+y_0+z_0}) + v_0(x_0 z_0 - e^{x_0+y_0+z_0}) + w_0(x_0 y_0 - e^{x_0+y_0+z_0})$$

where $u_0 = x_0^2 y_0^3 + z_0^4$, $v_0 = \sin x_0 - \cos y_0 \cos z_0$, $w_0 = x_0 y_0 z_0 - e^{x_0 + y_0 + z_0}$. The unit tangent to $r(t) = [e^{-t}, e^{-t}, -t - 20]$ is:

$$\mathbf{T} = \frac{1}{\sqrt{2e^{-2t} + 1}} [-e^{-t}, -e^{-t}, -1]$$

The tangential and normal accelerations are then:

$$a_T = \mathbf{a} \cdot \mathbf{T} = \frac{-e^{-t}a_1 - e^{-t}a_2 - a_3}{\sqrt{2e^{-2t} + 1}}$$

$$a_N = \sqrt{a_1^2 + a_2^2 + a_3^2 - a_T^2}$$

Problem Insight

The key realization is that the acceleration of a fluid particle is not simply $r''(t)$, we must use the chain rule which yields what is known as the material derivative $(\mathbf{v} \cdot \nabla)\mathbf{v}$, since the velocity field is defined over space and the particle moves through it. Students who differentiate $r(t)$ twice are computing the acceleration of the parametric curve, not the fluid acceleration. The curve $r(t)$ defines the path and the tangent vector provides the decomposition basis, but the acceleration must come from the velocity field itself.

Problem 6 - Solution

The curves r_2 and r_3 share the same x and z components, differing only in y , which ranges from $2 \cos t$ to 3. Together with r_1 , they sweep a surface over $t \in [-\pi/2, \pi/2]$. The volume between this surface and $f(r, \theta) = r - 2$, where $r = \sqrt{x^2 + y^2}$, is:

$$V = \int_{-\pi/2}^{\pi/2} \int_{2 \cos t}^3 \left(\sqrt{\sinh^2 t + y^2} - 2 - \sin t \right) dy dt$$

The inner integral is of the form $\int \sqrt{a^2 + y^2} dy$, which evaluates to:

$$\int_{2 \cos t}^3 \sqrt{\sinh^2 t + y^2} dy = \left[\frac{y}{2} \sqrt{\sinh^2 t + y^2} + \frac{\sinh^2 t}{2} \ln \left(y + \sqrt{\sinh^2 t + y^2} \right) \right]_{2 \cos t}^3$$

The resulting expression is intractable analytically. We therefore expand $\sinh t$ and $\cos t$ as Taylor series about $t = 0$:

$$\sinh t = \sum_{n=0}^{\infty} \frac{t^{2n+1}}{(2n+1)!}, \quad \cos t = \sum_{n=0}^{\infty} \frac{(-1)^n t^{2n}}{(2n)!}, \quad \sin t = \sum_{n=0}^{\infty} \frac{(-1)^n t^{2n+1}}{(2n+1)!}$$

Substituting and expanding $\sqrt{\sinh^2 t + y^2}$ using the binomial series $\sqrt{a^2 + b^2} = a\sqrt{1 + (b/a)^2} \approx a \sum_{n=0}^{\infty} \binom{1/2}{n} (b/a)^{2n}$ for $|b| > |a|$, the volume becomes:

$$V = \int_{-\pi/2}^{\pi/2} \int_{2 \cos t}^3 \left(\sum_{n=0}^{\infty} \binom{1/2}{n} \frac{\sinh^{2n} t}{y^{2n-1}} - 2 - \sum_{n=0}^{\infty} \frac{(-1)^n t^{2n+1}}{(2n+1)!} \right) dy dt$$

Integrating term by term over y from $2 \cos t$ to 3 and then over t from $-\pi/2$ to $\pi/2$, using the symmetry of even/odd terms:

$$V = \sum_{n=0}^{\infty} \binom{1/2}{n} \int_{-\pi/2}^{\pi/2} \frac{\sinh^{2n} t}{2n} (3^{2n} - (2 \cos t)^{2n}) dt - \int_{-\pi/2}^{\pi/2} (3 - 2 \cos t) \left(2 + \sum_{n=0}^{\infty} \frac{(-1)^n t^{2n+1}}{(2n+1)!} \right) dt$$

Each term in the series is integrable in closed form via the power series representations of $\sinh t$ and $\cos t$, giving the full approximation as an infinite series in π .

Problem Insight

The problem has two layers of difficulty. The first is geometric: recognizing that the three curves together sweep a surface rather than intersecting at discrete points. The second is analytic: the exact volume integral is intractable due to the $\sqrt{\sinh^2 t + y^2}$ term, and the student must recognize when to abandon exact computation in favor of systematic Taylor expansion. The binomial series expansion of the square root is the critical step, after which the integral decomposes into a sum of manageable terms. This is the essence of the Taylor series approximation method in applied mathematics knowing when the exact approach fails and how to construct a systematic approximation.

Problem 7 - Solution

To project the curve $r_1(t) = [\cos t, \sin t, 1]$ onto the plane $x - y - z = 1$ along the direction $\mathbf{d} = [1, 1, -1]$, we find s such that $P + s\mathbf{d}$ lies on the plane for each point $P = (\cos t, \sin t, 1)$:

$$(\cos t + s) - (\sin t + s) - (1 - s) = 1 \implies s = 2 - \cos t + \sin t$$

The projected curve is therefore:

$$\tilde{r}(t) = [\cos t + s, \sin t + s, 1 - s] = [2 + \sin t, 2 - \cos t + 2 \sin t, \cos t - \sin t - 1]$$

Differentiating:

$$\tilde{r}'(t) = [\cos t, \sin t + 2 \cos t, -\sin t - \cos t]$$

$$|\tilde{r}'(t)|^2 = \cos^2 t + (\sin t + 2 \cos t)^2 + (\sin t + \cos t)^2 = 4 + 3 \sin 2t + 2 \cos 2t$$

The arc length is:

$$L = \int_0^{2\pi} \sqrt{4 + 3 \sin 2t + 2 \cos 2t} dt = 2 \int_0^{2\pi} \sqrt{1 + \frac{3 \sin 2t + 2 \cos 2t}{4}} dt$$

Applying the binomial series $\sqrt{1+u} = \sum_{n=0}^{\infty} \binom{1/2}{n} u^n$ with $u = \frac{3 \sin 2t + 2 \cos 2t}{4}$ and integrating term by term:

$$L = 2 \int_0^{2\pi} \sum_{n=0}^{\infty} \binom{1/2}{n} \left(\frac{3 \sin 2t + 2 \cos 2t}{4} \right)^n dt$$

The $n = 0$ term contributes:

$$2 \int_0^{2\pi} dt = 4\pi$$

The $n = 1$ term vanishes by periodicity:

$$2 \cdot \frac{1}{2} \int_0^{2\pi} \frac{3 \sin 2t + 2 \cos 2t}{4} dt = 0$$

All odd powers of u vanish similarly by periodicity of $\sin 2t$ and $\cos 2t$ over $[0, 2\pi]$. The leading correction comes from $n = 2$:

$$\begin{aligned} 2 \cdot \left(-\frac{1}{8}\right) \int_0^{2\pi} \left(\frac{3 \sin 2t + 2 \cos 2t}{4}\right)^2 dt &= -\frac{1}{4} \int_0^{2\pi} \frac{9 \sin^2 2t + 12 \sin 2t \cos 2t + 4 \cos^2 2t}{16} dt \\ &= -\frac{1}{64} \int_0^{2\pi} (9 \sin^2 2t + 4 \cos^2 2t) dt = -\frac{1}{64} \cdot \pi(9 + 4) = -\frac{13\pi}{64} \end{aligned}$$

The arc length as a series is therefore:

$$L = 4\pi - \frac{13\pi}{64} + \sum_{n=3}^{\infty} \binom{1/2}{n} \cdot 2 \int_0^{2\pi} \left(\frac{3 \sin 2t + 2 \cos 2t}{4}\right)^n dt$$

with leading approximation $L \approx 4\pi - \frac{13\pi}{64}$.

Problem Insight

The projection step requires careful bookkeeping when finding s by substituting the parametric ray into the plane equation is straightforward but easy to botch. The arc length integral $\int \sqrt{4 + 3 \sin 2t + 2 \cos 2t} dt$ has no closed form, and the key insight is to factor out the constant term and apply the binomial series. The periodicity of $\sin 2t$ and $\cos 2t$ then kills all odd-order terms, so the series converges rapidly and the dominant approximation 4π is already quite accurate. This problem rewards students who recognize when a closed form is impossible and transition cleanly to a systematic series approximation.

Problem 8 - Solution

Suppose for contradiction that there exists a continuous vector field $\mathbf{F}$ on S^2 that is nowhere zero and always tangent to the sphere. Since $\mathbf{F}$ is nowhere zero, we can normalize it to a unit tangent vector field $\hat{\mathbf{F}} = \mathbf{F}/|\mathbf{F}|$ without any discontinuities. Geometrically, this is like having a head of hair on a sphere where every single hair lies perfectly flat against the surface, points in some direction, and no hair ever stands up or disappears.

We now construct a continuous deformation using $\hat{\mathbf{F}}$. For each point $\mathbf{r} \in S^2$, define:

$$H(\mathbf{r}, t) = \cos(\pi t) \mathbf{r} + \sin(\pi t) \hat{\mathbf{F}}(\mathbf{r}), \quad t \in [0, 1]$$

Since $\hat{\mathbf{F}}$ is tangent to the sphere, $\mathbf{r} \cdot \hat{\mathbf{F}} = 0$, so:

$$|H(\mathbf{r}, t)|^2 = \cos^2(\pi t) |\mathbf{r}|^2 + \sin^2(\pi t) |\hat{\mathbf{F}}|^2 = \cos^2(\pi t) + \sin^2(\pi t) = 1$$

meaning $H(\mathbf{r}, t)$ stays on S^2 for all t . Geometrically, H is sliding every point continuously along its hair direction, rotating it around the sphere. At $t = 0$:

$$H(\mathbf{r}, 0) = \mathbf{r}$$

every point is itself. At $t = 1$:

$$H(\mathbf{r}, 1) = -\mathbf{r}$$

every point has been sent to the point directly opposite it on the sphere — the point you would reach if you drilled straight through the center. This is called the *antipodal map*: it takes the north pole to the south pole, and every point (x, y, z) to $(-x, -y, -z)$. Imagine pressing your finger on a point on a balloon and pushing it all the way through to the other side — that is what the antipodal map does to every point simultaneously.

The function H is what mathematicians call a *homotopy* — a continuous deformation between two maps. Here H is a homotopy between the identity map (every point stays put) and the antipodal map (every point goes to its opposite). Geometrically, we have found a way to continuously slide every point of the sphere to its antipode without ever leaving the surface.

This is where the contradiction arises. Think about what it means to continuously deform one map into another on a sphere. The identity map preserves the orientation of the sphere — if you draw a small clockwise circle on the surface, it remains clockwise. The antipodal map on S^2 reverses orientation — that same clockwise circle becomes counterclockwise, as if the sphere were turned inside out. Geometrically, no continuous deformation can ever flip orientation: you cannot turn a right hand into a left hand by moving it smoothly through space without reflecting it. The mathematical tool that makes this precise is the *degree of a map*: the identity has degree $+1$ and

the antipodal map on S^2 has degree -1 , and homotopic maps must have the same degree.

Since H is a homotopy between two maps of different degree, we have a contradiction. Therefore our assumption was wrong: no continuous nowhere-zero tangent vector field can exist on S^2 .

Problem Insight

The hairy ball theorem is one of those results that is geometrically obvious once you see it you really cannot comb a hairy ball flat, but surprisingly difficult to prove rigorously without heavy machinery. The key insight is that a nowhere-zero tangent field gives you a way to continuously rotate every point to its antipode, and the antipodal map is fundamentally incompatible with the identity map on an even-dimensional sphere. Students should come away understanding that topology places hard constraints on what vector fields can look like globally, even when locally everything seems fine. Do not beat yourself up for not being able to do this problem, this problem is extremely difficult and is typically done in higher level analysis.

Problem 9 - Solution

Let $P = y \cos x$ and $Q = x + \sin x$. Computing the relevant partial derivatives:

$$\frac{\partial Q}{\partial x} = 1 + \cos x, \quad \frac{\partial P}{\partial y} = \cos x$$

So:

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1$$

By Green's theorem:

$$\oint_{\gamma} y \cos x \, dx + (x + \sin x) \, dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA = \iint_D dA = \pi ab$$

Problem Insight

The integral looks intimidating due to the $\sin x$ and $\cos x$ terms, which seem to suggest a difficult direct parametrization. The key insight is to recognize the integrand as a line integral of the form $\oint P \, dx + Q \, dy$ and immediately reach for Green's theorem. The curl $\partial Q/\partial x - \partial P/\partial y$ collapses to 1

exactly, reducing the double integral to the area of the ellipse. Students who attempt to parametrize the ellipse directly will find themselves in serious difficulty, the problem is designed to reward those who recognize the Green's theorem structure and think before they start writing!

Problem 10 - Solution

Part 1: Circulation

In polar coordinates, $d\mathbf{r} = dr \hat{r} + r d\theta \hat{\theta}$, so:

$$\mathbf{v} \cdot d\mathbf{r} = v_r dr + v_\theta r d\theta$$

On C : $r = a(1 + \varepsilon f(\theta))$, so $dr = a\varepsilon f'(\theta) d\theta$. The circulation is:

$$\Gamma_C = \oint_C v_r dr + v_\theta r d\theta = \int_0^{2\pi} [v_r \cdot a\varepsilon f'(\theta) + v_\theta \cdot r] d\theta$$

The v_r term is of order ε and vanishes at leading order. The v_θ term gives:

$$\Gamma_C = \int_0^{2\pi} \left(-U \left(1 + \frac{a^2}{r^2} \right) \sin \theta + \frac{\Gamma}{2\pi r} \right) r d\theta$$

The uniform flow terms contribute:

$$-U \int_0^{2\pi} (r + a^2/r) \sin \theta d\theta = 0$$

since $r = a(1 + \varepsilon f(\theta))$ and $\sin \theta$ is periodic with zero mean over $[0, 2\pi]$ for any smooth periodic $f(\theta)$. The circulation term contributes:

$$\int_0^{2\pi} \frac{\Gamma}{2\pi r} \cdot r d\theta = \frac{\Gamma}{2\pi} \int_0^{2\pi} d\theta = \Gamma$$

Therefore $\Gamma_C = \Gamma$ for any smooth periodic $f(\theta)$. The uniform flow terms integrate to zero because they involve products of periodic functions with zero mean — geometrically, the uniform flow contributes equal and opposite circulation on opposite sides of the contour, canceling exactly.

Problem Insight

The key insight is that the circulation is a topological quantity — it depends only on how many times the contour winds around the singularity at the origin, not on the exact shape of the contour. The uniform flow terms vanish by periodicity regardless of $f(\theta)$, leaving only the $\Gamma/2\pi r$ term which integrates to exactly Γ . Students should recognize this as an instance of the residue theorem in complex analysis, where only the $1/r$ term contributes to the contour integral.

Part 2: Scalar Vorticity and Stokes' Theorem

The scalar vorticity in polar coordinates is:

$$\omega_z = \frac{1}{r} \frac{\partial(rv_\theta)}{\partial r} - \frac{1}{r} \frac{\partial v_r}{\partial \theta}$$

Computing rv_θ :

$$rv_\theta = -Ur \left(1 + \frac{a^2}{r^2}\right) \sin \theta + \frac{\Gamma}{2\pi}$$
$$\frac{1}{r} \frac{\partial(rv_\theta)}{\partial r} = -\frac{U}{r} \left(1 - \frac{a^2}{r^2}\right) \sin \theta$$

Computing $\partial v_r / \partial \theta$:

$$v_r = U \left(1 - \frac{a^2}{r^2}\right) \cos \theta \implies \frac{1}{r} \frac{\partial v_r}{\partial \theta} = -\frac{U}{r} \left(1 - \frac{a^2}{r^2}\right) \sin \theta$$

Therefore:

$$\omega_z = -\frac{U}{r} \left(1 - \frac{a^2}{r^2}\right) \sin \theta + \frac{U}{r} \left(1 - \frac{a^2}{r^2}\right) \sin \theta = 0$$

The field is irrotational everywhere except at the origin, where the $\Gamma/2\pi r$ term introduces a singularity. Since the region D enclosed by C contains the origin, Stokes' theorem cannot be applied directly — the vorticity is not defined there. Geometrically, the singularity acts like a point vortex: the flow swirls around it but has zero local rotation everywhere else, analogous to how a whirlpool looks locally straight even though it curves globally. The corrected circulation formula is:

$$\oint_C \mathbf{v} \cdot d\mathbf{r} = \iint_D \omega_z dA + \Gamma = 0 + \Gamma = \Gamma$$

where Γ is the residual circulation contributed by the point vortex singularity at the origin, consistent with Part 1.

Problem Insight

The vorticity calculation looks like it should be nonzero given the complexity of the field, but everything cancels exactly — the field is a potential flow and is therefore irrotational away from the origin. The subtlety is that Stokes' theorem fails here not because the math is wrong but because the domain contains a singularity. Students must recognize that the $\Gamma/2\pi r$ term is not just another flow component, it is a topological feature of the field that cannot be removed by any smooth deformation of the contour, analogous to the $1/z$ term - which shows up rather often in Joukowski transforms - in complex analysis whose residue survives any contour integral enclosing the origin.

Part 3: Stream Function and Impermeability

We use the corrected stream function $\psi = U \left(r - \frac{a^2}{r} \right) \sin \theta + \frac{\Gamma}{2\pi} \ln r$. Computing $\partial\psi/\partial\theta$:

$$\frac{1}{r} \frac{\partial\psi}{\partial\theta} = \frac{1}{r} \cdot U \left(r - \frac{a^2}{r} \right) \cos \theta = U \left(1 - \frac{a^2}{r^2} \right) \cos \theta = v_r \checkmark$$

Computing $\partial\psi/\partial r$:

$$\begin{aligned} \frac{\partial\psi}{\partial r} &= U \left(1 + \frac{a^2}{r^2} \right) \sin \theta + \frac{\Gamma}{2\pi r} \\ -\frac{\partial\psi}{\partial r} &= -U \left(1 + \frac{a^2}{r^2} \right) \sin \theta - \frac{\Gamma}{2\pi r} = v_\theta \checkmark \end{aligned}$$

The impermeability condition requires $v_r = 0$ on C , i.e. at $r = a(1 + \varepsilon f(\theta))$:

$$v_r|_{r=a(1+\varepsilon f(\theta))} = U \left(1 - \frac{a^2}{a^2(1+\varepsilon f)^2} \right) \cos \theta = 0$$

Expanding to first order in ε using $(1 + \varepsilon f)^{-2} \approx 1 - 2\varepsilon f$:

$$U (1 - (1 - 2\varepsilon f)) \cos \theta = 2U\varepsilon f(\theta) \cos \theta = 0$$

For this to hold for all θ , we require:

$$f(\theta) \cos \theta = 0$$

This is the linear equation that $f(\theta)$ must satisfy to enforce impermeability to first order in ε .

Part 4: Pressure Coefficients and Lift

The pressure coefficient is:

$$C_p(\theta) = 1 - \frac{|v|^2}{U^2}$$

where $|v|^2 = v_r^2 + v_\theta^2$. On C at leading order ($r \approx a$):

$$v_r|_{r=a} = 0, \quad v_\theta|_{r=a} = -2U \sin \theta + \frac{\Gamma}{2\pi a}$$

So:

$$|v|^2|_{r=a} = \left(-2U \sin \theta + \frac{\Gamma}{2\pi a}\right)^2 = 4U^2 \sin^2 \theta - \frac{2U\Gamma \sin \theta}{\pi a} + \frac{\Gamma^2}{4\pi^2 a^2}$$

The pressure coefficient at leading order is:

$$C_p(\theta) = 1 - 4 \sin^2 \theta + \frac{2\Gamma \sin \theta}{\pi a U} - \frac{\Gamma^2}{4\pi^2 a^2 U^2}$$

To first order in ε , expanding v_r and v_θ at $r = a(1 + \varepsilon f)$:

$$v_r \approx 2U\varepsilon f(\theta) \cos \theta, \quad v_\theta \approx -2U \sin \theta + \frac{\Gamma}{2\pi a}(1 - \varepsilon f)$$

giving the first order correction to C_p :

$$C_p(\theta) \approx 1 - 4 \sin^2 \theta + \frac{2\Gamma \sin \theta}{\pi a U} - \frac{\Gamma^2}{4\pi^2 a^2 U^2} + \varepsilon \left(\frac{\Gamma f(\theta)}{\pi^2 a^2 U} - 4f(\theta) \sin \theta \cos \theta \right)$$

The total force on the body is:

$$\mathbf{F} = - \oint_C p \mathbf{n} dr$$

Using Bernoulli's equation $p = p_\infty + \frac{1}{2}\rho U^2 C_p$ and expanding the line element dr and outward normal $\mathbf{n}$ on C :

$$d\mathbf{r} = \left[a\varepsilon f'(\theta)\hat{r} + a(1 + \varepsilon f)\hat{\theta} \right] d\theta, \quad \mathbf{n} = \hat{r} - \varepsilon f'(\theta)\hat{\theta}$$

to first order in ε . The lift (force in $\hat{j}$ direction) is:

$$L = - \oint_C p \mathbf{n} \cdot \hat{j} dr = -\frac{1}{2} \rho U^2 \oint_C C_p (\sin \theta - \varepsilon f'(\theta) \cos \theta) a(1 + \varepsilon f) d\theta$$

At leading order, using $\int_0^{2\pi} \sin^2 \theta d\theta = \pi$ and $\int_0^{2\pi} \sin \theta d\theta = 0$:

$$L^{(0)} = -\frac{1}{2} \rho U^2 a \int_0^{2\pi} \left(1 - 4 \sin^2 \theta + \frac{2\Gamma \sin \theta}{\pi a U} \right) \sin \theta d\theta = \rho U \Gamma$$

This is the *Kutta-Joukowski theorem*: the lift per unit span equals $\rho U \Gamma$, depending only on the free stream velocity and circulation, not on the shape of the body. The first order correction in ε gives:

$$L^{(1)} = -\frac{1}{2} \rho U^2 a \int_0^{2\pi} \varepsilon \left(f(\theta) C_p^{(0)} \sin \theta - f'(\theta) C_p^{(0)} \cos \theta \right) d\theta$$

so the total lift to first order is:

$$L = \rho U \Gamma + L^{(1)}$$

Problem Insight

Problem 10 is a complete derivation of classical potential flow theory around a cylinder. The key results are: (1) circulation is a topological invariant that depends only on the winding of the contour around the singularity; (2) the flow is irrotational everywhere except at the origin, making Stokes' theorem fail without a correction term; (3) the impermeability condition linearizes cleanly in ε ; and (4) the Kutta-Joukowski theorem $L = \rho U \Gamma$ emerges naturally from the pressure integral, showing that lift is entirely determined by circulation. Students should appreciate that this entire framework such as potential flow, stream functions, pressure coefficients, and lift, follows from a single velocity field and the tools of vector calculus.

Problem 11 - Solution 1

This problem is fundamentally about finding the shortest path between two points constrained to lie on a time-varying surface. Since the surface changes with time, we treat all geometric quantities as functions of time t .

At any fixed instant t , the Mars surface is given by:

$$z = f(x, y, t) = m(t) \cos(n(t)x) + n(t) \sin(m(t)y)$$

We define:

$$m_t = m(t), \quad n_t = n(t)$$

so that the surface at time t becomes:

$$z = f_t(x, y) = m_t \cos(n_t x) + n_t \sin(m_t y)$$

The two locations (houses) must lie on this surface at each time t , so we write:

$$X(t) = (x_0, y_0, f_t(x_0, y_0)), \quad Y(t) = (x_1, y_1, f_t(x_1, y_1))$$

Explicitly:

$$z_0(t) = m(t) \cos(n(t)x_0) + n(t) \sin(m(t)y_0)$$

$$z_1(t) = m(t) \cos(n(t)x_1) + n(t) \sin(m(t)y_1)$$

Any path constrained to the surface can be written using two functions:

$$x(u), \quad y(u), \quad u \in [0, 1]$$

and the surface constraint gives:

$$z(u, t) = f_t(x(u), y(u))$$

So the path is:

$$\mathbf{r}(u, t) = (x(u), y(u), f_t(x(u), y(u)))$$

The differential arc length is given by:

$$ds^2 = dx^2 + dy^2 + dz^2$$

Since $z = f_t(x, y)$, we use:

$$dz = f_x(x, y, t) dx + f_y(x, y, t) dy$$

We compute the partial derivatives:

$$f_x = -m(t)n(t) \sin(n(t)x), \quad f_y = m(t)n(t) \cos(m(t)y)$$

Substituting into the arc length expression:

$$ds = \sqrt{dx^2 + dy^2 + (f_x dx + f_y dy)^2}$$

Thus, the total walking distance along a given path becomes:

$$L(t) = \int_0^1 \sqrt{x'(u)^2 + y'(u)^2 + \left(-m(t)n(t) \sin(n(t)x)x'(u) + m(t)n(t) \cos(m(t)y)y'(u) \right)^2} du$$

The minimum distance required to travel between the two houses at time t is therefore:

$$D(t) = \min_{x(u), y(u)} L(t)$$

Problem Insight

This problem tests the ability to translate a time-dependent geometric surface into a path optimization problem. The key idea is that at each fixed time t , the surface is effectively static, even though its shape depends on time through $m(t)$ and $n(t)$. A crucial insight is that distance on a surface is not measured in straight lines in $\mathbb{R}^3$, but along curves constrained to the surface. This forces us to express the path in terms of two free functions $x(u)$ and $y(u)$, while the third coordinate is determined by the surface constraint.

The expression for arc length comes directly from the 3D Pythagorean theorem, combined with the chain rule applied to $z = f(x, y, t)$. The resulting functional $L(t)$ depends on the chosen path, and the true solution is the path that minimizes it.

In practice, this type of problem is typically solved numerically, since closed-form geodesics are rarely obtainable for nonlinear time-varying surfaces such as this one.

Problem 11 - Solution 2

The surface is:

$$M = \{(x, y, z) \in \mathbb{R}^3 \mid z = m(t) \cos(n(t)x) + n(t) \sin(m(t)y)\}$$

where $m(t)$ and $n(t)$ are twice differentiable functions of time. We seek the minimum distance path between two points $X = (x_0, y_0, z_0)$ and $Y = (x_1, y_1, z_1)$ on M , i.e. the geodesic.

A curve $\gamma(s) = (x(s), y(s), z(s))$ on M satisfies $z(s) = m \cos(nx) + n \sin(my)$ at every point. Its arc length is:

$$L[\gamma] = \int_0^1 \sqrt{\dot{x}^2 + \dot{y}^2 + \dot{z}^2} ds$$

Since z is determined by x and y on M , we compute:

$$\dot{z} = \frac{\partial z}{\partial x} \dot{x} + \frac{\partial z}{\partial y} \dot{y} = -mn \sin(nx) \dot{x} + mn \cos(my) \dot{y}$$

Substituting:

$$L[\gamma] = \int_0^1 \sqrt{(1 + m^2 n^2 \sin^2(nx)) \dot{x}^2 - 2m^2 n^2 \sin(nx) \cos(my) \dot{x} \dot{y} + (1 + m^2 n^2 \cos^2(my)) \dot{y}^2} ds$$

This defines the metric tensor g_{ij} on M :

$$g_{xx} = 1 + m^2 n^2 \sin^2(nx), \quad g_{xy} = -m^2 n^2 \sin(nx) \cos(my), \quad g_{yy} = 1 + m^2 n^2 \cos^2(my)$$

The geodesic equations are obtained by minimizing $L[\gamma]$ via the Euler-Lagrange equations. For a curve parametrized by arc length, these give:

$$\ddot{x}^k + \Gamma_{ij}^k \dot{x}^i \dot{x}^j = 0, \quad k = 1, 2$$

where Γ_{ij}^k are the Christoffel symbols:

$$\Gamma_{ij}^k = \frac{1}{2} g^{kl} \left(\frac{\partial g_{il}}{\partial x^j} + \frac{\partial g_{jl}}{\partial x^i} - \frac{\partial g_{ij}}{\partial x^l} \right)$$

For our metric, the nonzero Christoffel symbols are:

$$\Gamma_{xx}^x = \frac{-m^3 n^3 \sin(nx) \cos(nx)}{1 + m^2 n^2 \sin^2(nx)}, \quad \Gamma_{yy}^y = \frac{m^3 n^3 \sin(my) \cos(my)}{1 + m^2 n^2 \cos^2(my)}$$

giving the geodesic equations:

$$\ddot{x} - \frac{m^3 n^3 \sin(nx) \cos(nx)}{1 + m^2 n^2 \sin^2(nx)} \dot{x}^2 = 0$$

$$\ddot{y} + \frac{m^3 n^3 \sin(my) \cos(my)}{1 + m^2 n^2 \cos^2(my)} \dot{y}^2 = 0$$

These are nonlinear second order ODEs with no closed form solution for arbitrary $m(t)$ and $n(t)$.

The minimum distance between X and Y is therefore:

$$d(X, Y) = \int_0^1 \sqrt{g_{ij} \dot{x}^i \dot{x}^j} ds$$

evaluated along the solution of the geodesic equations subject to boundary conditions $\gamma(0) = X$ and $\gamma(1) = Y$. This must in general be solved numerically for given $m(t)$, $n(t)$, X , and Y .

Problem Insight

This problem has no closed form solution in general and that is precisely the point. The surface M changes over time, making the geodesic a moving target. The key insight is to recognize that finding the shortest path on a curved surface is a variational problem: minimize the arc length functional subject to the constraint that the path lies on M . This leads naturally to the geodesic equations via Euler-Lagrange, which are the curved-space analogue of Newton's second law. The metric tensor g_{ij} encodes how distances are distorted by the curvature of M , and the Christoffel symbols encode how the coordinate basis vectors change as you move along the surface. Students should appreciate that even a seemingly simple surface can produce intractable geodesic equations, and that numerical methods are often the only recourse in practice.

Problem 12 - Solution

Part 1: Airfoil Profile and Pressure Distributions

The airfoil surfaces satisfy $y'' + 2y' + 5y = 0$ with characteristic equation:

$$r^2 + 2r + 5 = 0 \implies r = -1 \pm 2i$$

giving general solution $y(x) = e^{-x}(C_1 \cos 2x + C_2 \sin 2x)$.

For the upper surface, $y_U(0) = 0$ gives $C_1 = 0$, and $y'_U(0) = 0.8$ gives $2C_2 = 0.8$, so $C_2 = 0.4$:

$$y_U(x) = 0.4e^{-x} \sin 2x$$

For the lower surface, $y_L(0) = 0$ gives $C_1 = 0$, and $y'_L(0) = -0.4$ gives $2C_2 = -0.4$, so $C_2 = -0.2$:

$$y_L(x) = -0.2e^{-x} \sin 2x$$

The leading edge is at $x = 0$ where $y_U = y_L = 0$. The trailing edge occurs where $y_U = y_L$, i.e. $0.4e^{-x} \sin 2x = -0.2e^{-x} \sin 2x$, giving $\sin 2x = 0$, so $x = \pi/2$. The chord length (leading to trailing edge distance) is:

$$c = \frac{\pi}{2}$$

The radius of curvature of the airfoil surfaces is:

$$\kappa(x) = \frac{|y''|}{(1 + y'^2)^{3/2}}$$

For the upper surface:

$$y'_U = 0.4e^{-x}(2 \cos 2x - \sin 2x), \quad y''_U = 0.4e^{-x}(-4 \sin 2x - 2 \cos 2x - 2 \cos 2x + \sin 2x)$$

$$y''_U = 0.4e^{-x}(-3 \sin 2x - 4 \cos 2x)$$

giving radius of curvature:

$$R_U(x) = \frac{(1 + 0.16e^{-2x}(2\cos 2x - \sin 2x)^2)^{3/2}}{0.4e^{-x}|3\sin 2x + 4\cos 2x|}$$

and similarly for the lower surface with the factor 0.2 replacing 0.4.

For the upper surface pressure, $p''_U + 2.4p'_U + 2.44p_U = 0$ has characteristic equation:

$$r^2 + 2.4r + 2.44 = 0 \implies r = -1.2 \pm i$$

giving general solution $p_U(y) = e^{-1.2y}(C_1 \cos y + C_2 \sin y)$.

Applying $p_U(0) = 0$ gives $C_1 = 0$, and $p'_U(0) = 4$ gives $C_2 = 4$:

$$p_U(y) = 4e^{-1.2y} \sin y$$

For the lower surface pressure, $p''_L + 4p'_L + 5p_L = 0$ has characteristic equation:

$$r^2 + 4r + 5 = 0 \implies r = -2 \pm i$$

giving general solution $p_L(x) = e^{-2x}(C_1 \cos x + C_2 \sin x)$.

Applying $p_L(0) = 0$ gives $C_1 = 0$, and $p'_L(0) = -1$ gives $C_2 = -1$:

$$p_L(x) = -e^{-2x} \sin x$$

Part 2: Lift and Drag

The contour Y is formed by $y_U(x)$ and $y_L(x)$ over $x \in [0, \pi/2]$. The outward normal to the upper surface is:

$$\mathbf{n}_U = \frac{[-y'_U, 1]}{\sqrt{1 + y'^2_U}}, \quad \mathbf{n}_L = \frac{[y'_L, -1]}{\sqrt{1 + y'^2_L}}$$

The lift and drag are:

$$L = \oint_Y p \mathbf{n} \cdot \hat{i} \, dr = \int_0^{\pi/2} \left(\frac{-p_U y'_U}{\sqrt{1 + y'^2_U}} - \frac{-p_L y'_L}{\sqrt{1 + y'^2_L}} \right) \sqrt{1 + y'^2} \, dx$$

$$= \int_0^{\pi/2} (-p_U y'_U + p_L y'_L) dx$$

Substituting $p_U(y_U(x))$ and $p_L(y_L(x))$, using $y_U = 0.4e^{-x} \sin 2x$ and $y_L = -0.2e^{-x} \sin 2x$:

$$L = \int_0^{\pi/2} \left(-4e^{-1.2(0.4e^{-x} \sin 2x)} \sin(0.4e^{-x} \sin 2x) \cdot 0.4e^{-x} (2 \cos 2x - \sin 2x) \right. \\ \left. + (-e^{-2(-0.2e^{-x} \sin 2x)} \sin(-0.2e^{-x} \sin 2x)) \cdot (-0.2e^{-x}) (2 \cos 2x - \sin 2x) \right) dx$$

Similarly for drag:

$$D = \int_0^{\pi/2} (p_U - p_L) dx = \int_0^{\pi/2} (4e^{-1.2y_U} \sin y_U + e^{-2y_L} \sin(-y_L)) dx$$

These integrals have no closed form and must be evaluated numerically or via Taylor expansion for small surface deflections.

Part 3: Circulation over a Cardioid

The cardioid in Cartesian coordinates is $(x^2 + y^2 - x)^2 = 8((x + 2)^2 + y^2)$. Converting to polar coordinates $x = r \cos \theta$, $y = r \sin \theta$:

$$(r^2 - r \cos \theta)^2 = 8(r \cos \theta + 2)^2 + 8r^2 \sin^2 \theta$$

The velocity field is $\mathbf{v} = [s^2, -2e^{-s} \sin t - 3]$ for $t > 0$, $s \in \mathbb{R}$. The circulation is:

$$\Gamma = \oint_C \mathbf{v} \cdot d\mathbf{r} = \oint_C s^2 ds + (-2e^{-s} \sin t - 3) dt$$

Applying Green's theorem over the region D enclosed by the cardioid:

$$\Gamma = \iint_D \left(\frac{\partial}{\partial s} (-2e^{-s} \sin t - 3) - \frac{\partial}{\partial t} (s^2) \right) dA = \iint_D 2e^{-s} \sin t dA$$

Expanding e^{-s} and $\sin t$ as Taylor series:

$$e^{-s} = \sum_{n=0}^{\infty} \frac{(-s)^n}{n!}, \quad \sin t = \sum_{n=0}^{\infty} \frac{(-1)^n t^{2n+1}}{(2n+1)!}$$

giving:

$$\Gamma = 2 \iint_D \left(\sum_{n=0}^{\infty} \frac{(-s)^n}{n!} \right) \left(\sum_{n=0}^{\infty} \frac{(-1)^n t^{2n+1}}{(2n+1)!} \right) dA$$

The area of the cardioid $(x^2 + y^2 - x)^2 = 8((x+2)^2 + y^2)$ is computed by finding its polar form. Expanding and collecting terms gives $r = 1 + \cos \theta$ up to a scaling factor, whose enclosed area is $\frac{3\pi}{2}$ per unit. The leading order term of the circulation is:

$$\Gamma \approx 2 \iint_D t dA$$

which by symmetry of the cardioid region evaluates to a series in the moments of the cardioid area.

Part 4: Velocity Field and Circulation

By Bernoulli's principle, $p \propto 1/|v|^2$ with proportionality factor 2.2, so:

$$|v|^2 = \frac{2.2}{p}$$

For the upper surface, using $p_U = 4e^{-1.2y} \sin y$:

$$|v_U| = \sqrt{\frac{2.2}{4e^{-1.2y} \sin y}} = \sqrt{\frac{0.55e^{1.2y}}{\sin y}}$$

Expanding to second order Taylor polynomial around $y = 0$:

$$e^{1.2y} \approx 1 + 1.2y + 0.72y^2, \quad \sin y \approx y - \frac{y^3}{6} \approx y$$

$$|v_U| \approx \sqrt{\frac{0.55(1 + 1.2y + 0.72y^2)}{y}}$$

Similarly for the lower surface using $p_L = -e^{-2x} \sin x$:

$$|v_L| = \sqrt{\frac{2.2}{e^{-2x} \sin x}} \approx \sqrt{\frac{2.2e^{2x}}{x}}$$

expanding $e^{2x} \approx 1 + 2x + 2x^2$ to second order. The circulation of the velocity field around the airfoil contour Y is:

$$\begin{aligned}\Gamma &= \oint_Y \mathbf{v} \cdot d\mathbf{r} = \int_0^{\pi/2} (|v_U| - |v_L|) dx \\ &\approx \int_0^{\pi/2} \left(\sqrt{\frac{0.55(1 + 1.2x + 0.72x^2)}{x}} - \sqrt{\frac{2.2(1 + 2x + 2x^2)}{x}} \right) dx\end{aligned}$$

Part 5: Angle of Attack

For an angle of attack $\alpha = 10^\circ$, the velocity field is rotated by α :

$$\begin{aligned}\mathbf{v}_\alpha &= R(\alpha)\mathbf{v} = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \\ &= \begin{pmatrix} v_1 \cos 10^\circ - v_2 \sin 10^\circ \\ v_1 \sin 10^\circ + v_2 \cos 10^\circ \end{pmatrix}\end{aligned}$$

The straight line from the trailing edge $(\pi/2, 0)$ to the cardioid boundary divides the region into upper U and lower L parts. The circulations are:

$$\Gamma_U = \oint_U \mathbf{v}_\alpha \cdot d\mathbf{r}, \quad \Gamma_L = \oint_L \mathbf{v}_\alpha \cdot d\mathbf{r}$$

Applying Green's theorem to each region:

$$\Gamma_U = \iint_{D_U} \left(\frac{\partial v_{\alpha,2}}{\partial x} - \frac{\partial v_{\alpha,1}}{\partial y} \right) dA, \quad \Gamma_L = \iint_{D_L} \left(\frac{\partial v_{\alpha,2}}{\partial x} - \frac{\partial v_{\alpha,1}}{\partial y} \right) dA$$

Since the rotation matrix $R(\alpha)$ is constant, the curl of $\mathbf{v}_\alpha$ equals the curl of $\mathbf{v}$ rotated by α :

$$\frac{\partial v_{\alpha,2}}{\partial x} - \frac{\partial v_{\alpha,1}}{\partial y} = \cos \alpha \left(\frac{\partial v_2}{\partial x} - \frac{\partial v_1}{\partial y} \right)$$

Using the velocity field from Part 4 and expanding the sinusoidal components to second order Taylor polynomials, the circulations become:

$$\Gamma_U = \cos(10^\circ) \iint_{D_U} \left(\frac{\partial v_2}{\partial x} - \frac{\partial v_1}{\partial y} \right) dA$$

$$\Gamma_L = \cos(10^\circ) \iint_{D_L} \left(\frac{\partial v_2}{\partial x} - \frac{\partial v_1}{\partial y} \right) dA$$

with $\Gamma_U + \Gamma_L = \cos(10^\circ) \cdot \Gamma$ where Γ is the total circulation from Part 4.

Problem Insight

Problem 12 builds a complete aerodynamic pipeline from scratch using only ODEs and vector calculus. The airfoil profile emerges from a damped oscillator equation, giving physically realistic exponentially decaying sinusoidal surfaces. The pressure distributions follow similar ODEs with different damping, and Bernoulli's principle connects pressure to velocity. The key insight across all parts is that the same mathematical tools such as Green's theorem, Taylor series, and contour integrals appear repeatedly at every stage of the analysis. The angle of attack calculation shows that a rigid rotation of the velocity field scales the circulation by $\cos \alpha$, a result with direct physical meaning: lift decreases as the airfoil pitches beyond its optimal angle. Students who complete this problem have essentially derived the foundations of thin airfoil theory from first principles.